

# Aaryan Singh

Motivated student with a proven track record of academic excellence, leadership capabilities, and community involvement.

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**GitHub:** <https://github.com/aaryan2k>

## EXPERIENCE

### FiPet, Vernon Hills (Remote) — Intern

Jan 2026 - Present

- Developed and shipped frontend features using React Native, improving app usability and overall user experience.
- Collaborated with a startup team to integrate Firebase-backed data with the UI, gaining hands-on experience in building production-ready mobile interfaces.

### Xperience Consulting, Vernon Hills — Board of Directors

Aug 2023 - Present

- Led coding camps for children, transforming them into a nationally recognized non-profit.
- Secured funding and gained business management experience through project leadership.

### Code Ninjas, Libertyville — Code Sensei

Nov 2021 - Aug 2024

- Collaborated with fellow instructors to refine teaching methods.
- Taught game creation to children, emphasizing creativity and problem-solving.

### Tachyon Dynamics, Vernon Hills (Remote) — Intern

July 2023 - Aug 2023

- Developed a testing framework using Scapy and Wireshark to validate IPv6 connectivity and monitor network traffic

## EDUCATION

### University of Illinois at Urbana-Champaign

Pursuing a Computer Science Major and a Data Science Minor.

**GPA:** 3.8/4.00

**Coursework:** Data Structures, Numerical Methods I, Computer Architecture, Data Science Discovery, Data Science Exploration

**Clubs:** Illini Electric Motorsports (IEM), Project: Code, Fantasy Trading Room, Sports Analytics Society, Illini WBB Analytics Team

### Vernon Hills High School

July 2020 - May 2024

**GPA:** 4.388/4 (Weighted) 3.9/4 (Unweighted)

Seven07

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## SKILLS

Communication.

Leadership.

Problem Solving.

Teamwork.

## TECHNICAL SKILLS

**Programming Languages:** Java, JavaScript, Python, C++, C#, SQL

**Web & Frontend:** HTML, CSS, React, Tailwind

**Machine Learning & Data:** scikit-learn, statsmodels, TensorFlow, LangGraph, pandas, NumPy

**Tools & Platforms:** GitHub, GitHub Copilot, Streamlit, PyQt5, Python-CAN, Matplotlib

## AWARDS

- AP Scholar with Distinction
- 2023 ST3AM Showcase Winner
- Chicago Metro History Day Regional Qualifier
- Vernon Hills High School Athletic Award
- Illinois State Music Teachers Association Achievement in Music Level 12
- National Merit Commended Student
- Fall 2024 Dean's List

## LANGUAGES

English, Hindi, Spanish

**Coursework:** Computer Science A, Data Structures (IIT'), Advanced Topics in Computer Science

**Clubs:** Computer Science Club, Robotics Club, National Honor Society, AI Committee, Tennis Team

## PROJECTS

### **HoopsNet**

Implemented a React frontend and Python data pipeline that cleaned and analyzed NBA datasets, engineered weighted statistical features, and trained TensorFlow models to predict NBA champions, with outputs visualized in the application.

### **Illini Women's Basketball Analytics Team**

Applied data analysis using Python and pandas to transform CSV game data into visual insights, collaborating with coaching staff to define requirements and developing lineup-level metrics such as plus-minus and efficiency to support data-driven decision making.

### **Linear Regression Analysis (Data Science Project)**

Applied statistical modeling using Python and statsmodels to analyze an NFL passing dataset in a team setting, performing feature scaling and LASSO-based feature selection, evaluating model performance with  $R^2$  and RMSE, and interpreting model coefficients to explain predictive relationships.

### **LabSpimbot Competition (Computer Architecture Project)**

Developed a team-based MIPS SPIMBot to compete in a simulated tournament, implementing playpen locking/unlocking, real-time interrupt handling, and strategic sabotage avoidance. Programmed autonomous behavior to score points by managing bunnies, responding to interrupts, and coordinating offensive and defensive actions to meet qualification requirements across multiple map scenarios.